



Bahria University, Islamabad
Department of Software Engineering

Machine Learning Lab
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Teacher: Engr. Kamran Khan

Student Name: Faisal khan

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Task No:	Task Wise Marks		Documentation Marks		Total Marks (20)
	Assigned	Obtained	Assigned	Obtained	
1	3		5		
2	3				
3	3				
4	3				
5	3				

Comments:

Signature

Write a Python program to implement Simple Linear Regression

[3]
✓ Os

```
import numpy as np

# X = Independent Variable (e.g., Years of Experience)
# Y = Dependent Variable (e.g., Salary)
X = np.array([1.1, 1.5, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, 9.0])
Y = np.array([39343, 37731, 43525, 60150, 57189, 83988, 93691, 102000, 113931, 116969])

n = len(X)
print(f"Dataset loaded with {n} data points.")
```

Dataset loaded with 10 data points.

[4]
✓ Os

```
# Step 2: Manual Calculation of Regression Coefficients
sum_x = np.sum(X)
sum_y = np.sum(Y)
sum_xy = np.sum(X * Y)
sum_x_sq = np.sum(X ** 2)

# Calculate Slope (b1)
# b1 = (n * sum(xy) - sum(x) * sum(y)) / (n * sum(x^2) - (sum(x))^2)
b1 = (n * sum_xy - sum_x * sum_y) / (n * sum_x_sq - (sum_x ** 2))

# Calculate Intercept (b0)
# b0 = (sum(y) - b1 * sum(x)) / n
b0 = (sum_y - b1 * sum_x) / n

print(f"Calculated Intercept (b0): {b0:.4f}")
print(f"Calculated Slope (b1): {b1:.4f}")
```

... Calculated Intercept (b0): 23926.3273
Calculated Slope (b1): 10928.1916

[5]
✓ Os

```
# Step 3: Making Predictions

# 1. Predictions on the training data points
Y_pred = b0 + b1 * X

# 2. Making predictions on new, unseen data points (e.g., 2.5, 6.5, and 8.5 years of experience)
new_X = np.array([2.5, 6.5, 8.5])
new_Y_pred = b0 + b1 * new_X

for x_val, y_val in zip(new_X, new_Y_pred):
    print(f"Predicted Y for X = {x_val} (Experience): {y_val:.2f}")
```

Predicted Y for X = 2.5 (Experience): 51246.81
Predicted Y for X = 6.5 (Experience): 94959.57
Predicted Y for X = 8.5 (Experience): 116815.96

[6]
✓ Os

```
# Step 4: Model Evaluation (MSE and R-squared)

# 1. Mean Squared Error (MSE): Average of squared residuals[cite: 1]
mse = np.mean((Y - Y_pred) ** 2)

# 2. R-squared (r^2) Coefficient of Determination[cite: 1]
sum_y_sq = np.sum(Y ** 2)
numerator_r = n * sum_xy - sum_x * sum_y
denominator_r = np.sqrt((n * sum_x_sq - sum_x ** 2) * (n * sum_y_sq - sum_y ** 2))
r = numerator_r / denominator_r
r_squared = r ** 2

print(f"Mean Squared Error (MSE): {mse:.4f}")
print(f"R-squared (r^2): {r_squared:.4f}")
```

Mean Squared Error (MSE): 22868599.4976
R-squared (r^2): 0.9735

[7]
✓ Os

```
# Step 5: Visualization of the Regression Line and Data Points
plt.figure(figsize=(8, 5))

# Plot actual data points as a scatter plot
plt.scatter(X, Y, color='red', label='Actual Data Points')

# Plot the predicted regression line
plt.plot(X, Y_pred, color='blue', linewidth=2, label='Regression Line (Best Fit)')

plt.title('Simple Linear Regression: Salary vs Experience')
plt.xlabel('Independent Variable (X - Years of Experience)')
plt.ylabel('Dependent Variable (Y - Salary)')
plt.legend()
plt.grid(True, linestyle='--', alpha=0.5)
plt.show()
```



Thank you
